Shraeyaa Dhaigude **D20A - 22**

BC Experiment - 10

Aim: Explore the Cryptocurrency Landscape

Theory:

This experiment explores the **cryptocurrency landscape** by combining real-world market data analysis with blockchain technology research. The focus asset for this analysis is **Chainlink (LINK)**, and data was collected from **three reputable sources** as per the lab requirement:

1. **CoinGecko API** – Provided historical price data, market capitalization, and trading volumes in a structured JSON format, enabling direct processing in Python.
2. **CoinMarketCap** – Offered real-time and historical price charts, circulating supply, and market rank for visual cross-verification.
3. **Crypto.com** – Supplied quick-access market statistics, simplified price trends, and cryptocurrency news updates.

Cryptocurrency Landscape

1. **Definition** – Cryptocurrencies are decentralized digital assets built on blockchain networks, enabling secure, transparent, and immutable transactions without central authority.
2. **Market Diversity** – Thousands of cryptocurrencies exist, each serving unique roles:
   - **Store of Value** – Bitcoin, Litecoin.
   - **Utility Tokens** – Chainlink, Ethereum.
   - **Stablecoins** – USDT, USDC.
   - **Governance Tokens** – UNI, AAVE.
3. **Volatility** – Prices are highly sensitive to market sentiment, technological developments, and regulatory updates.
4. **Data Analysis Importance** – Tracking historical trends, market capitalization, and trading volume is essential for understanding asset performance.
5. **Advancements** – Growing use of **DeFi platforms**, **NFT ecosystems**, **layer-2 scaling**, and **cross-chain protocols**.

Data Collection Process

The program used the **CoinGecko API** to automate historical data fetching for Chainlink. The retrieved data was processed with **Pandas** to compute:

- **Daily Returns** – Percentage change between consecutive days to measure volatility.
- **Moving Averages (MA7, MA30)** – Short-term and long-term trend indicators.
- **Cumulative Returns** – Growth factor over time.
- **Rolling Volatility** – Short-term risk measurement based on daily return fluctuations.

Visualization & Dashboard

Using **Matplotlib**, a six-panel dashboard was created to display:

1. **Price Trend** – Daily price movement over the selected period.
2. **Price vs Volume** – Correlation between trading activity and price changes.
3. **Daily Returns Distribution** – Statistical spread of daily gains/losses.

4. **Moving Averages** – Technical analysis trend lines for price movement.
5. **Cumulative Returns** – Overall growth performance.
6. **Rolling Volatility** – Market risk trends over time.

Technological Analysis

Beyond market data, the study covered **advancements in blockchain technology**, including:

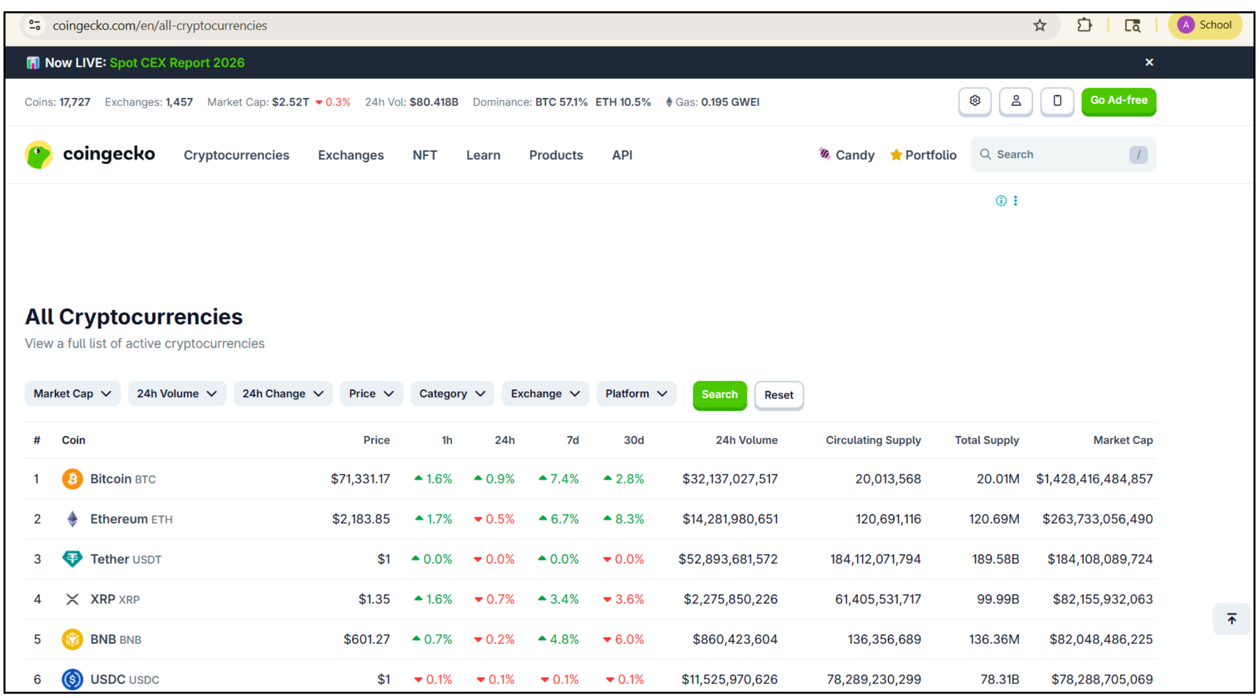
- **Consensus Mechanisms** – PoW (Proof of Work), PoS (Proof of Stake), and hybrid systems.
- **Scalability Solutions** – Layer-2 protocols, sharding, and sidechains to increase throughput.
- **Interoperability Protocols** – Chainlink's **CCIP**, Polkadot, and Cosmos enabling cross-chain data exchange.
- **Privacy Features** – Zero-knowledge proofs (zk-SNARKs) and ring signatures to ensure transaction confidentiality.

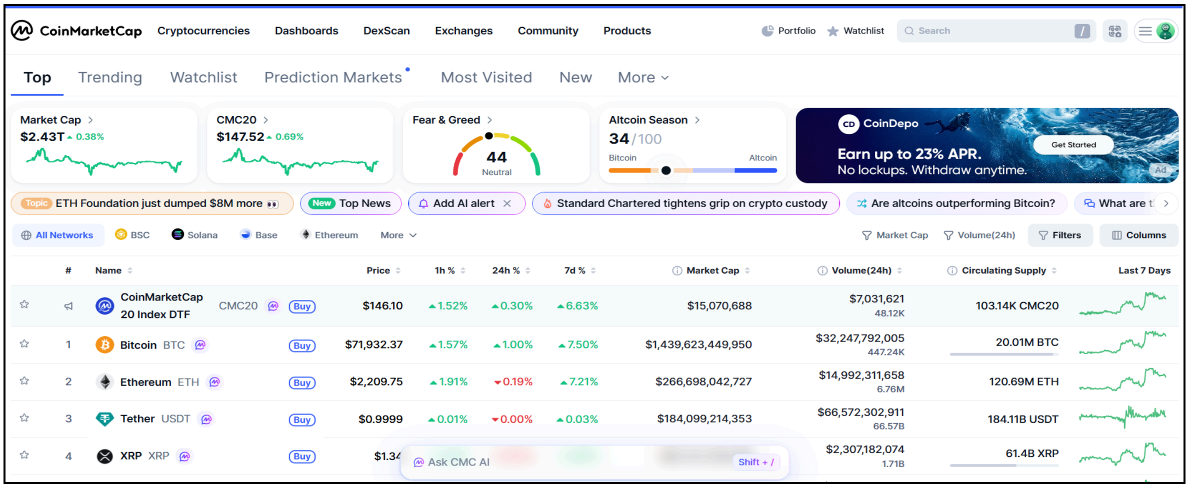
Chainlink's **roadmap** highlights staking implementation, advanced oracle features, and enhanced cross-chain capabilities, aligning with industry trends toward **decentralization, interoperability, and security**.

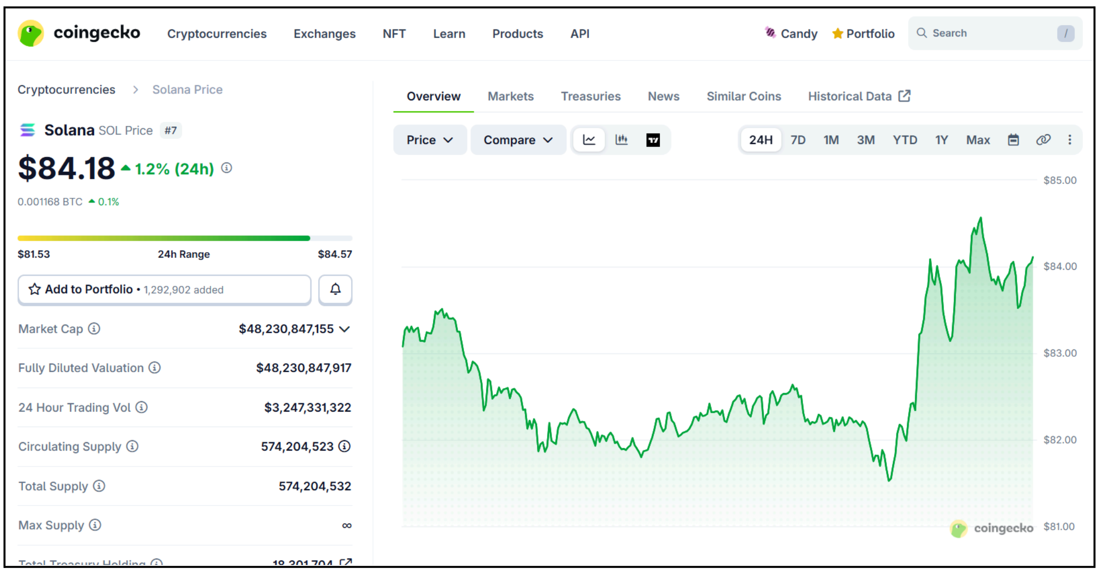
Market Trends & Adoption

- **Growing Merchant Acceptance** – More businesses accepting cryptocurrency payments.
- **Wallet Growth** – Increasing downloads and installations of crypto wallets.
- **Network Activity** – Higher transaction volumes and active addresses, signaling strong adoption.
- **Integration with Traditional Finance** – Partnerships between crypto platforms and fintech/payment providers.
- **Regulatory Developments** – Countries exploring frameworks for safe cryptocurrency adoption.

Implementation




CoinMarketCap
Cryptocurrencies
Dashboards
DexScan
Exchanges
Community
Products
Portfolio
Watchlist
Search
Top
Trending
Watchlist
Prediction Markets
Most Visited
New
More
Market Cap
$2.43T 0.38%
CMC20
$147.52 0.69%
Fear & Greed
44
Neutral
Altcoin Season
34/100
Bitcoin
Altcoin
CoinDepo
Earn up to 23% APR.
No lockups. Withdraw anytime.
Get Started
ETH Foundation just dumped $8M more
Top News
Add AI alert
Standard Chartered tightens grip on crypto custody
Are altcoins outperforming Bitcoin?
All Networks
BSC
Solana
Base
Ethereum
More
Market Cap
Volume(24h)
Filters
Columns
Name Price 1h % 24h % 7d % Market Cap Volume(24h) Circulating Supply Last 7 Days
CoinMarketCap 20 Index DTF CMC20 Buy $146.10 1.52% 0.30% 6.63% $15,070,688 $7,031,621 48.12K 103.14K CMC20
1 Bitcoin BTC Buy $71,932.37 1.57% 1.00% 7.50% $1,439,623,449,950 $32,247,792,005 447.24K 20.01M BTC
2 Ethereum ETH Buy $2,209.75 1.91% 0.19% 7.21% $266,698,042,727 $14,992,311,658 6.76M 120.69M ETH
3 Tether USDT Buy $0.9999 0.01% 0.00% 0.03% $184,099,214,353 $66,572,302,911 66.57B 184.11B USDT
4 XRP XRP Buy $1.34 $2,307,182,074 1.71B 61.4B XRP
Ask CMC AI
Shift + /


coingecko
Cryptocurrencies
Exchanges
NFT
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Search
Cryptocurrencies > Solana Price
Solana SOL Price #7
$84.18 1.2% (24h)
0.001168 BTC 0.1%
$81.53 24h Range $84.57
Add to Portfolio • 1,292,902 added
Market Cap $48,230,847,155
Fully Diluted Valuation $48,230,847,917
24 Hour Trading Vol $3,247,331,322
Circulating Supply 574,204,523
Total Supply 574,204,532
Max Supply ∞
Overview
Markets
Treasuries
News
Similar Coins
Historical Data
Price
Compare
24H
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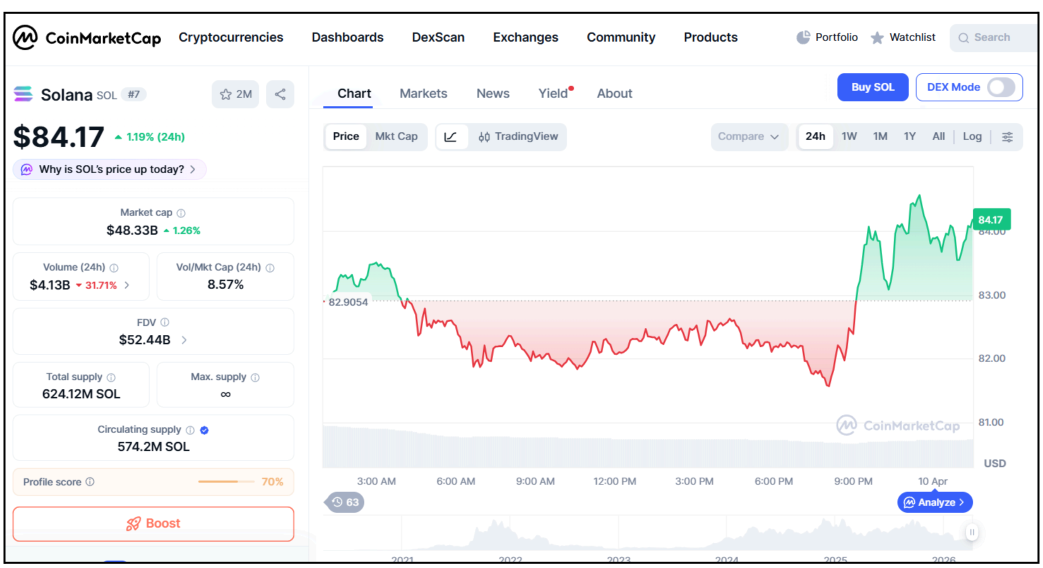
CoinMarketCap
Cryptocurrencies
Dashboards
DexScan
Exchanges
Community
Products
Portfolio
Watchlist
Search
Solana SOL #7
2M
$84.17 1.19% (24h)
Why is SOL's price up today?
Market cap
$48.33B 1.26%
Volume (24h)
$4.13B 31.71%
Vol/Mkt Cap (24h)
8.57%
FDV
$52.44B
Total supply
624.12M SOL
Max. supply
∞
Circulating supply
574.2M SOL
Profile score
70%
Boost
Chart
Markets
News
Yield
About
Buy SOL
DEX Mode
Price
Mkt Cap
TradingView
Compare
24h
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10 Apr
63
Analyze
2021
2022
2023
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Utilizing Api for analysis and visualization :
Code:

```
import requests

import matplotlib.pyplot as plt

import pandas as pd


# ---------- Fetch Data ----------

def get_historical_data(coin_id, days='180', currency='usd'):

    url = f'https://api.coingecko.com/api/v3/coins/{coin_id}/market_chart'

    params = {'vs_currency': currency, 'days': days}

    response = requests.get(url, params=params)

    return response.json()


# ---------- Dashboard Function ----------

def ethereum_dashboard(days='180'):

    coin_id = 'ethereum'

    data = get_historical_data(coin_id, days)


    # Create DataFrame

    prices_df = pd.DataFrame(data['prices'], columns=['timestamp', 'price'])

    volumes_df = pd.DataFrame(data['total_volumes'], columns=['timestamp', 'volume'])


    prices_df['timestamp'] = pd.to_datetime(prices_df['timestamp'], unit='ms')

    volumes_df['timestamp'] = pd.to_datetime(volumes_df['timestamp'], unit='ms')


    df = pd.merge(prices_df, volumes_df, on='timestamp')

    df.set_index('timestamp', inplace=True)


    # Calculate extra metrics

    df['returns'] = df['price'].pct_change()
```

```
df['MA7'] = df['price'].rolling(window=7).mean()
df['MA30'] = df['price'].rolling(window=30).mean()
df['cumulative'] = (1 + df['returns']).cumprod()
df['volatility'] = df['returns'].rolling(window=7).std() * 100

# ---------- Plot Dashboard ----------
fig, axes = plt.subplots(3, 2, figsize=(16, 12))
fig.suptitle(f'{coin_id.capitalize()} Dashboard (Last {days} Days)', fontsize=16, fontweight='bold')

# 1. Price Trend
axes[0, 0].plot(df.index, df['price'], color='blue')
axes[0, 0].set_title('Price Trend')
axes[0, 0].set_ylabel('Price (USD)')
axes[0, 0].set_xlabel('Year-Month') # Add xlabel
axes[0, 0].grid(True)

# 2. Price vs Volume
axes[0, 1].plot(df.index, df['price'], color='blue', label='Price')
axes[0, 1].bar(df.index, df['volume'], color='orange', alpha=0.3, label='Volume')
axes[0, 1].set_title('Price vs Volume')
axes[0, 1].set_xlabel('Year-Month') # Add xlabel
axes[0, 1].set_ylabel('Price (USD) / Volume') # Add ylabel
axes[0, 1].legend()
axes[0, 1].grid(True)

# 3. Daily Returns Distribution
axes[1, 0].hist(df['returns'].dropna(), bins=50, alpha=0.7, color='purple')
axes[1, 0].set_title('Daily Returns Distribution')
axes[1, 0].set_xlabel('Daily Return')
axes[1, 0].set_ylabel('Frequency')
```

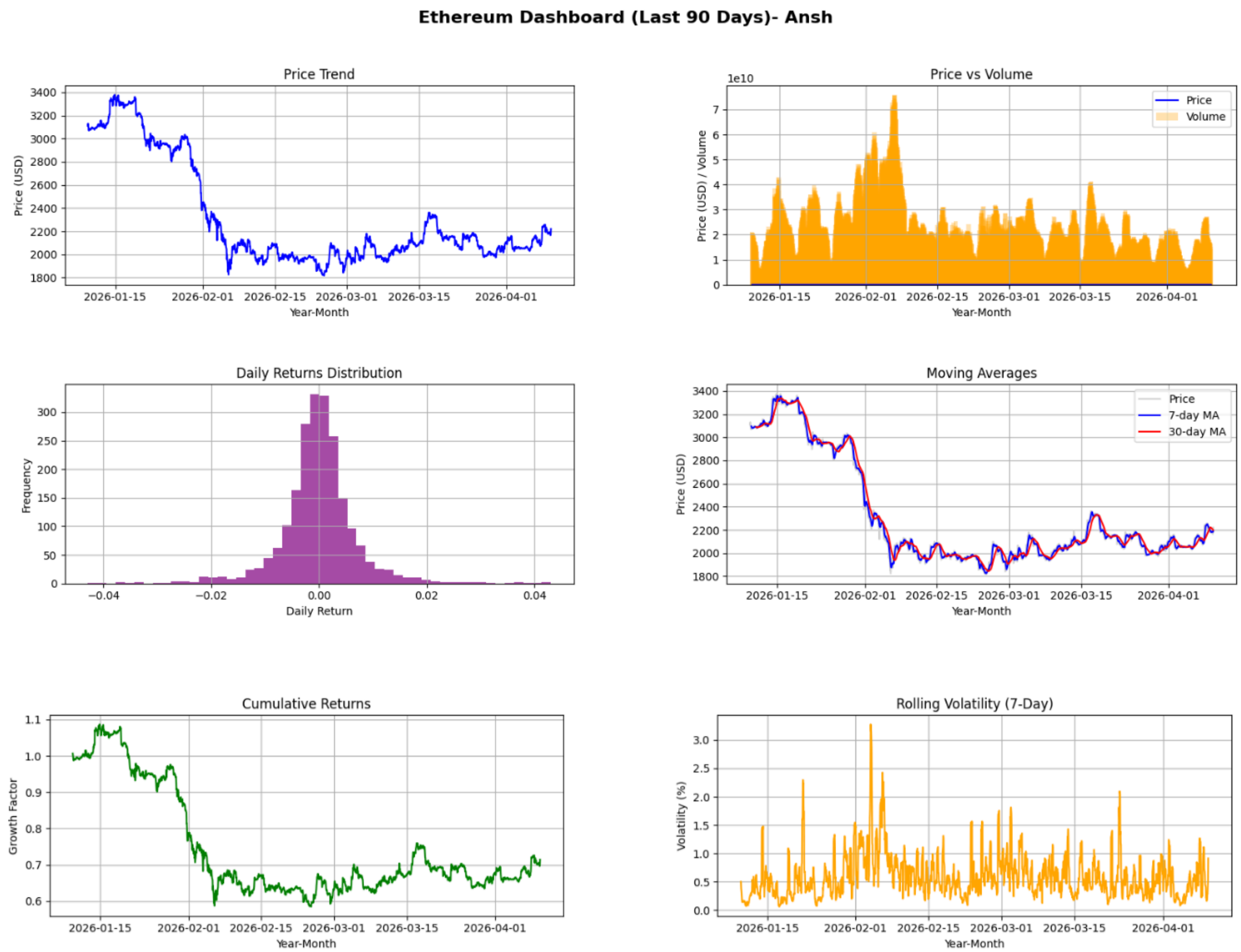
```python
    axes[1, 0].grid(True)

    # 4. Moving Averages
    axes[1, 1].plot(df.index, df['price'], color='lightgray', label='Price')
    axes[1, 1].plot(df.index, df['MA7'], color='blue', label='7-day MA')
    axes[1, 1].plot(df.index, df['MA30'], color='red', label='30-day MA')
    axes[1, 1].set_title('Moving Averages')
    axes[1, 1].set_xlabel('Year-Month')
    axes[1, 1].set_ylabel('Price (USD)')
    axes[1, 1].legend()
    axes[1, 1].grid(True)

    # 5. Cumulative Returns
    axes[2, 0].plot(df.index, df['cumulative'], color='green')
    axes[2, 0].set_title('Cumulative Returns')
    axes[2, 0].set_ylabel('Growth Factor')
    axes[2, 0].set_xlabel('Year-Month')
    axes[2, 0].grid(True)
    # 6. Volatility
    axes[2, 1].plot(df.index, df['volatility'], color='orange')
    axes[2, 1].set_title('Rolling Volatility (7-Day)')
    axes[2, 1].set_ylabel('Volatility (%)')
    axes[2, 1].set_xlabel('Year-Month')
    axes[2, 1].grid(True)

    plt.tight_layout(rect=[0, 0, 1, 0.96])
    plt.subplots_adjust(hspace=0.5, wspace=0.3) # Adjust vertical and horizontal spacing
    plt.show()
# ---------- Run Dashboard ----------
ethereum_dashboard('90')
```

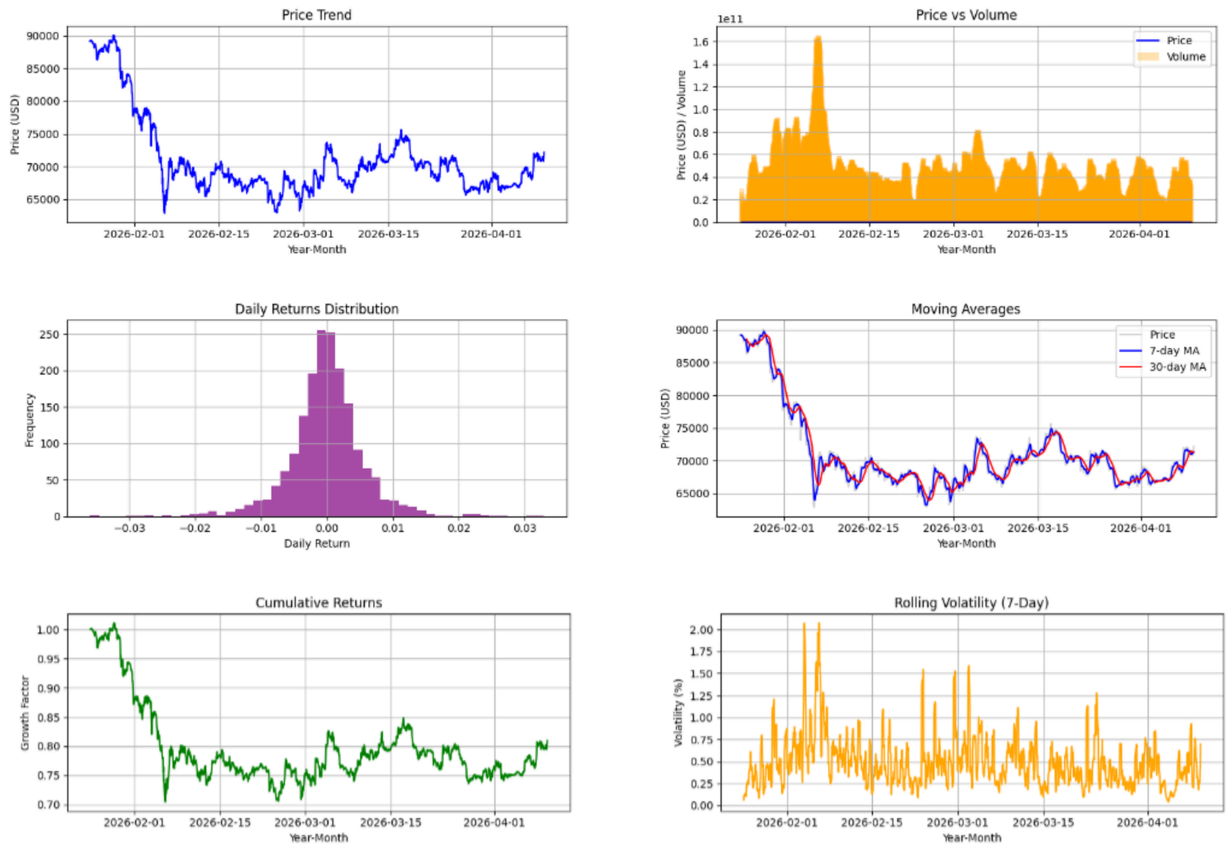
Output:



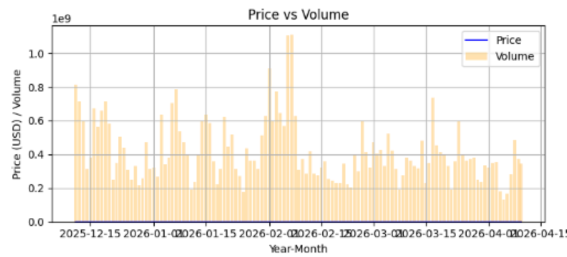
Conclusion:

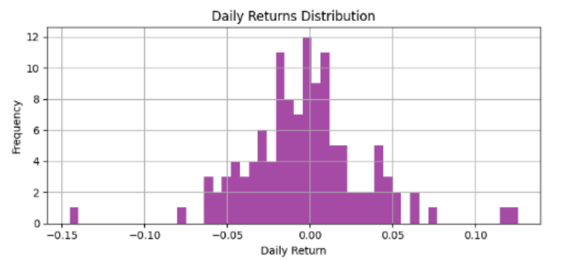
By combining automated market data processing, visual trend analysis, blockchain technology evaluation, and adoption metrics, this experiment presents a holistic view of the cryptocurrency ecosystem. It demonstrates how blockchain's technical foundations connect with real-world market behavior, providing insights into both current trends and future advancements in the digital asset industry.

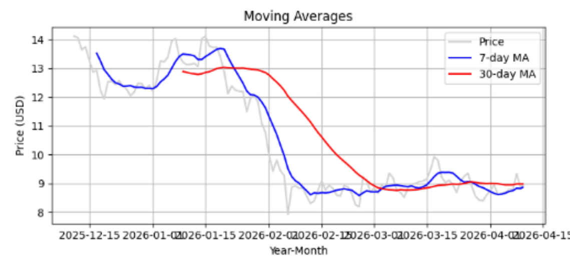
Bitcoin Dashboard (Last 75 Days)- Prajjwal

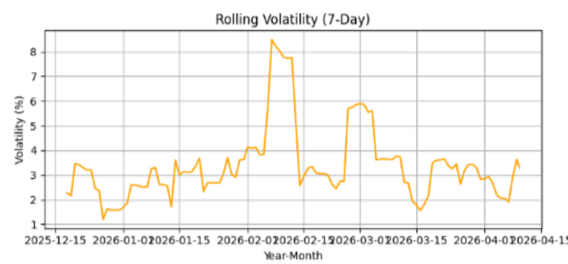


Chainlink Dashboard (Last 120 Days)-Mahvish









Solana Dashboard (Last 60 Days)- Shraeyaa